A Binaural Audio Dataset for Misophonia Trigger Sounds

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Abstract

Misophonia is characterized by strong negative emotional and physiological reactions to specific everyday sounds, yet assistive audio technologies capable of selectively suppressing these triggers remain limited.

In this work, we present the first large-scale, open-access binaural dataset specifically designed to support misophonia-focused selective Active Noise Control (ANC). The dataset comprises audio mixtures with triggering or non-triggering (control) foreground sounds and background sounds. These are mixed with spatialization using Binaural Room Impulse Response (BRIR) data from the SADIE II database [1] to produce acoustic conditions sufficiently realistic to generalize to real-world use cases. Trigger sounds are organized using a clinically inspired taxonomy, while control and background sounds are chosen from categories not commonly recognized as triggering. The dataset is designed to be readily extensible to incorporate any source dataset of sounds, currently using sounds from FSD50K [2], ESC-50 [3], and FOAMS [4]. We release both a fixed canonical dataset of 30,000 binaural mixtures and an extensible Python pipeline that enables scalable, on-the-fly mixture generation.

We also specify and implement an online validation protocol in which participants with self-reported misophonia rate discomfort for paired trigger and control mixtures. Data collection has not yet been conducted due to pending ethics approval, but the full protocol and analysis plan are defined a priori. The planned analysis employs crossed subject–by–item mixed-effects models to estimate both global and personalized trigger effects

Together, the dataset, generation pipeline, and validation framework provide foundational resources for developing and benchmarking selective, patient-centric ANC systems aimed at reducing the everyday burden of misophonia.

1 Introduction

Misophonia is a disorder characterized by extreme negative emotional and physiological reactions to specific everyday sounds, such as chewing, throat clearing, or pen clicking [5, 6]. Although not yet recognized as a formal diagnosis in the DSM-5-TR or ICD-11, growing clinical evidence shows that misophonia can severely impair occupational, social, and academic functioning [6, 7]. Emotional responses commonly include anger, anxiety, and disgust, while physiological reactions involve increased heart rate, body tension, and somatic discomfort [7]. Its prevalence among the the general population is still debated. Different studies estimate the incidence of misophonia between 3% and 20% [7–9].

Despite growing recognition, assistive audio technologies tailored to misophonia remain underdeveloped. Suffers' lives could be meaningfully improved by giving them control over unwanted trigger sounds. One concrete approach is selective Active Noise Control (ANC), in which triggering sounds are suppressed while non-triggering sounds are preserved, allowing users to remain engaged with their surroundings without exposure to the specific sounds they find aversive.

This paper presents the first large-scale, open-access binaural dataset specifically constructed to support selective ANC models tailored to misophonia triggers. It consists of audio mixtures in which both triggering and non-triggering foreground sounds are combined with background sounds. These mixtures are rendered binaurally with spatialization, yielding realistic acoustic conditions for training and evaluation and thus well suited to developing systems intended to operate in everyday, real-time scenarios.

In addition, we plan an online validation survey in which participants with misophonia rate discomfort for paired trigger/control mixtures. The experiment and analysis has been implemented, but due to not being able to obtain ethics approval in time for the deadline of this report, we have not yet conducted the experiment. Given this, we have opted to outline the experiment and the subsequent planned analysis in this report. Even though we have not been able to test it, our a priori expectation is that trigger mixtures will elicit higher discomfort.

Taken together, this work makes three contributions. First, we propose a framework for misophonia-focused ANC that couples taxonomy, data generation, and evaluation. Second, we curate and release a large-scale, open-access binaural corpus of mixed trigger and control scenes, alongside an extensible pipeline for on-the-fly mixture generation. Third, we design a statistically grounded online validation experiment to quantify trigger–control discomfort differences in screened participants, providing a template for future empirical evaluation once ethics approval allows data collection.

Our Python framework for generating the sound mixtures is released on GitHub[1]. This includes a small dummy dataset with a few mixtures, isolated trigger sounds and associated metadata.

The remainder of this report is organized as follows. Section 2 outlines prior work on misophonia, selective audio control, and binaural datasets. Section 3 explains how we construct trigger and non-trigger binaural audio mixtures. Section 4 outlines our planned online validation study. Section 5 situates our dataset within the current state of research, discussing the direction of future research. Section 6 summarizes the main contributions.

2 Related Works

Previous research has addressed several ingredients of misophonia-focused ANC, but typically in isolation. Some studies show that trigger-like sounds can be classified using machine learning, enabling non-selective ANC to be activated automatically; others curate and validate trigger stimuli for use in clinical research; and others demonstrate real-time, spatially aware selective suppression for non-misophonia sound classes. However, no existing effort combines all required components: trigger definitions informed by clinical research, large-scale datasets suitable for machine learning, and

[1] https://github.com/LukasFT/misophonia-dataset

binaural realism reflecting everyday listening conditions. In this section, we review prior work on misophonia triggers, selective sound control, and binaural audio datasets to clarify what is already feasible and to motivate the contributions of this report.

The feasibility of *classifying* misophonia-related sounds have been shown in previous literature. Bahmei et al. [10] evaluated a Vision Transformer for classifying misophonia triggers from short audio segments, reporting high accuracy and specificity. Wunrow [11] developed a CNN-based "selective noise canceling" prototype and observed reduced self-reported reactions to canceled triggers in an online survey. Together, these studies demonstrate that trigger recognition is tractable. However, they rely on general-purpose audio datasets from which misophonia-relevant classes were retrospectively selected without validation and relation to the clinical literature. And their code and preprocessing pipelines are not publicly available, limiting reproducibility and further development of their methods. Moreover, both approaches operate on non-spatial clips and focus on detection or global suppression rather than controllable, category-level extraction and suppression in realistic auditory contexts.

The closest prior system to our goal of *selectively* suppressing only some sounds while passing others, is, to our knowledge, Bahmei [12]. They suppressed sounds that some subjects with Autism Spectrum Disorder found unpleasant, and evaluated the efficacy with a small pilot study. The pipeline is organized into two stages. In the first stage, a convolutional–recurrent classifier trained on UrbanSound8K [13] detects aversive environmental sounds such as sirens, jackhammers, and dog barks. In the second stage, a ratio-mask denoiser based on a deep neural network – trained with LibriSpeech [14] and ESC-50 [3] – attenuates the detected sources. Although the system provides a convincing proof of concept, it remains preliminary relative to our misophonia-oriented framework. The evaluation was conducted with a small ($N = 8$) group of autistic participants who exhibited decreased sound tolerance but not clinically defined misophonia. The target classes were not validated as misophonia triggers or aligned with clinical taxonomies. Moreover, the analysis relied on descriptive averages without inferential statistics, treating participants and items as fixed, limiting generalizability. The design neither modeled nor preserved binaural spatial cues. Finally, the use of cloud-based processing introduced substantial latency, limiting real-world, real-time use.

Fortunately, recent work has shown the feasibility of ANC targeting only specific sounds in real time. Namely, Veluri et al. [15] introduced *Semantic Hearing*, a real-time system that extracts or suppresses predefined sound classes (e.g., alarms, car horns) while preserving spatial cues using a transformer-based network trained on synthesized mixtures. However, they did not target common misophonia trigger categories (e.g., chewing, slurping) nor evaluate with misophonia-validated stimuli. Nonetheless, their dataset curation and sound supression model inspire a large part of our work.

A source of such misophonia-validated stimuli is FOAMS [4], which provides 50 short audio clips of trigger sounds. It is mainly intended to lower stimulus-preparation barriers in psychology studies.

They obtained discomfort ratings from a pilot study of people with self-reported misophonia based on their responses to The Duke Misophonia Questionnaire (DMQ) [16]. It offers a good, validate resource that can be used as a reference, as well as a taxonomy that can be used across machine learning and medical research. Similarly, at the same time we were developing our dataset, Oh et al. [17] released a dataset with validated, audio-visual misophonia trigger stimuli. They relied on a more extensive, hierarchical taxonomy described in Williams et al. [18]. However, clips from these datasets are isolated (no background), non-binaural, and small in scale for machine learning applications.

The aforementioned Duke Misophonia Questionnaire (DMQ) [16] provides a validated and clinically grounded assessment of misophonia that fits well with the aims of our dataset and validation study. The DMQ captures the key dimensions emphasized in the clinical literature through dedicated subscales developed using large-sample psychometric analyses. These subscales are trigger frequency, affective, physiological, and cognitive responses, coping behaviors across phases of exposure, impairment, and maladaptive beliefs. Importantly, the instrument also defines a composite *Symptom Severity* score by combining the affective, physiological, and cognitive subscales. This score offers a concise, reliable index of core misophonia symptomatology.

In summary, existing research demonstrates that several components of misophonia-focused ANC are individually feasible, but none integrates all parts of our aim: clinically grounded trigger definitions, selective sound extraction, and spatial realism that generalize to real-time use cases. Prior work shows that trigger classification is achievable [10, 11], that aversive environmental sounds can be detected and attenuated using learned masks [12], and that real-time selective extraction with preserved spatial cues is possible for some audio classes [15]. Clinical resources such as FOAMS and the DMQ further supply validated trigger exemplars and standardized assessment tools [4, 16]. Yet these contributions remain fragmented. To address this gap, our dataset provides trigger and non-trigger binaural audio clips grounded in clinical taxonomy, enabling selective misophonia-focused ANC in realistic auditory settings.

3 Binaural Dataset Curation

A main contributions of this report is the curation of a large-scale, open-source, binaural dataset of misophonia-centric audio. This dataset serves as the foundation for an ANC model designed to identify and suppress triggering sounds. Because the model operates directly on human auditory input, it is essential that both the input and output signals be binaural. To reduce the risk of overfitting to overly simple or atypical auditory scenes, we aim to generate realistic mixtures. Our approach takes an isolated misophonia trigger (or control) sound and combines it with arbitrary background audio—provided the background contains neither triggers nor control sounds—then binaurally mixes them. The resulting complex audio scenes contain a trigger (or control) event embedded within a spatially localized acoustic environment.

As noted, our binaural mixtures are derived from existing audio data. In addition to releasing a fixed *canonical* binaural misophonia dataset, we also release a parallelizable pipeline capable of handling an arbitrary number of source datasets and generating binaural mixtures on-the-fly. Similar in spirit to PyTorch's `DataLoader`, our methodology is well suited to the high data throughput required by deep learning models and enables future researchers to incorporate additional datasets with minimal friction. Furthermore, the pipeline supports straightforward data augmentation through manipulation of various mixing hyperparameters (see Section 3.3). In this section, we introduce our consolidated taxonomy for organizing the audio data, describe the preparation of the source datasets, and detail our binaural mixing procedure.

3.1 Taxonomy Preparation

The binaural mixes always contain one or more background sounds and either (a) one or more trigger sounds or (b) one or more control sounds. In the trigger case, we label the foreground using the trigger taxonomy and the background using the background taxonomy. In the control case, we label the foreground using the control taxonomy and the background using the same background taxonomy as before. In other words, trigger, control, and background sounds are each drawn from their own separate label sets, which we describe in the following subsections.

3.1.1 Trigger Taxonomy

To establish our trigger taxonomy, we reviewed the literature to identify the most frequently reported trigger sounds as well as the taxonomies used in existing datasets. Three taxonomies were particularly relevant. First, the FOAMS dataset [4] provides 50 misophonia triggers grouped into 10 classes, which have been empirically validated in clinical research [19]. These classes include chewing gum, human breathing, swallowing, clearing throat, plastic crumpling, newspaper flipping, water drops, typing, knife cutting, and basketball dribbling, and are organized within a broader taxonomy modeled after UrbanSound8K [13]. Second, Claiborn et al. [20] compiled a list of triggers from a literature survey. Their classes largely align with FOAMS, but they additionally include mechanical sounds such as pen clicking and pencil tapping, and broaden chewing gum to chewing. Finally, we examined the taxonomy introduced in the DMQ [18], shown in Fig. 1.

The DMQ taxonomy is extensive and clearly structured, which led us to adopt it as our overarching taxonomy. We retain a primary focus, however, on the FOAMS trigger classes, with the modification of broadening chewing gum to chewing. Accordingly, our final trigger set includes chewing, human breathing, swallowing, clearing throat, plastic crumpling, newspaper flipping, water drops, typing, knife cutting, and basketball dribbling. The added classes of Claiborn et al. [20] do not appear in the dataset as we were unable to gather any samples. But, by adopting the DMQ framework, these trigger categories and others can be seamlessly incorporated into the taxonomy and, by extension, into our full pipeline.

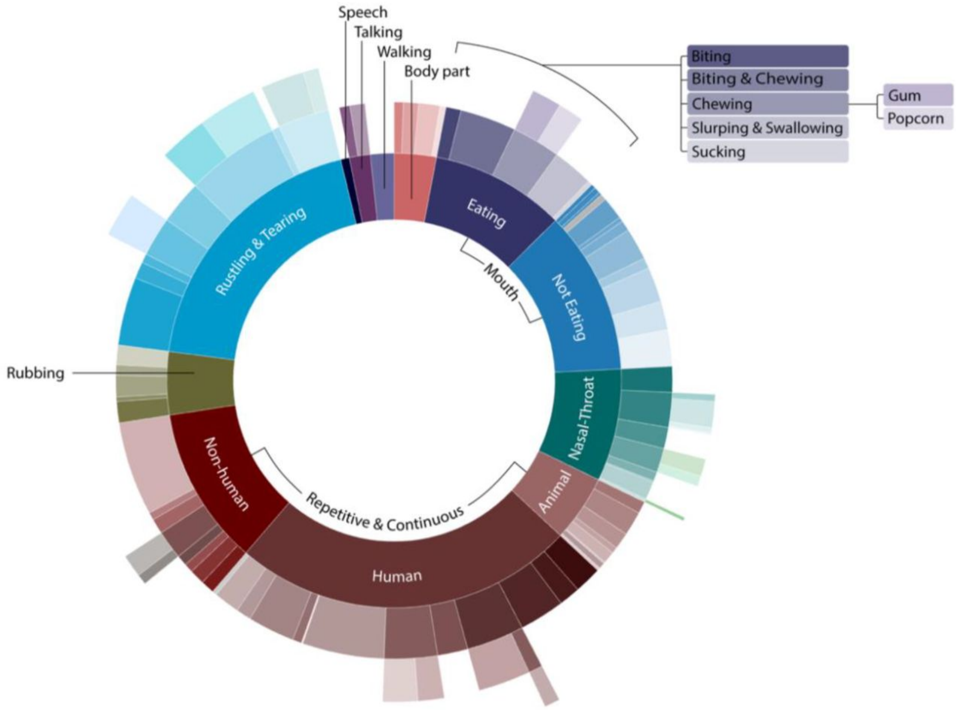

Figure 1: Misophonia Trigger sound taxonomy, as outlined by Williams et al. [18]. Visual originally from Oh et al. [17]

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3.1.2 Control and Background Taxonomy

Of equal importance to identifying trigger classes is defining appropriate control classes. In their empirical validation of the FOAMS misophonia classes, Hansen et al. [19] compared triggering responses against a control baseline consisting of coffee shop ambiance, harp playing, and hairdryer use. These sounds were selected because “they involve human action but are not commonly reported as a misophonic trigger,” reflecting the fact that most (though not all) misophonia triggers arise from human actions. For our purposes, we excluded coffee shop ambiance from the control category. Because our framework requires mutually exclusive trigger, control, and background classes—and because we aim to generate realistic audio scenes—we determined that coffee shop ambiance is better suited as background audio. To diversify the control set, we added two additional classes, chosen according to the criteria of Hansen et al.: piano playing and vacuuming. Importantly, neither sound appears as a common trigger in any of the misophonia studies we surveyed.

For background sounds, our goal was to compile classes that exclude both our control sounds and any audio identified as a misophonia trigger in the literature. As a starting point, we examined the

632 unique classes in the AudioSet Ontology [21]. Because manually evaluating each class would be impractical, we used ChatGPT 5 to filter out all classes resembling either our control sounds or any of the 12 trigger classes defined in our taxonomy. We then instructed it to retain only "neutral" or "positive" sound classes—those unlikely to annoy an average listener. This process yielded 192 candidate classes. As a final safeguard, we manually reviewed these classes and removed any associated with known misophonia triggers, such as clock ticking or finger snapping. This resulted in a verified set of 184 background classes.

Although mutually exclusive, both the control and background taxonomies are subsets of the AudioSet Ontology.

3.2 Source Datasets

To obtain trigger, control, and background sounds, we gathered samples from the FOAMS, ESC-50, and FSD50K datasts. As mentioned, the FOAMS dataset, developed in 2023 from the UrbanSound8K dataset, contains 50 misophonia trigger sounds across 10 classes. Each sound clip is a high quality, isolated example of a unique trigger class [4]. The ESC-50 dataset for environmental sound classification, first released in 2015, contains 2000 labeled recordings divided equally between 50 classes [3]. The classes are loosely grouped into five major categories: animal sounds, natural soundscapes and water sounds, human (non-speech) sounds, domestic sounds, and urban noises. Each audio file from ESC-50 is in a 5-second-long format. The FSD50K dataset, the largest of the three, is a collection of over 50 thousand human-labeled sound files from the Freesound archive [2]. Sound clip durations range from .3 to 30 seconds. Labels were manually added using over 300 classes from the AudioSet taxonomy. FSD50K is split into development (training) and evaluation (test) sets, of approximate size 40k and 10k respectively. Labels in the development set are correct but occasionally incomplete, whereas evaluation set labels are both correct and exhaustive [2].

3.2.1 Sound Event Selection

Given that our trigger taxonomy is an expansion of the FOAMS trigger classes, we collect all samples of the FOAMS dataset. However, both ESC-50 and FSD50K (i.e., FSD50K eval and FSD50K dev) datasets contain sound events of classes which are irrelevant, or even contrary to our needs. We require sound events that belong to, and only to, the trigger, control, or background classes listed in the respective taxonomies. Moreover, as some sound events in FSD50K are multi-labeled [2], we require that any multi-labeled sound clip does not contain classes across our three taxonomies. In order to filter out sound events according to these two requirements, we developed mappings from the classes of ESC-50 and FSD50K to the trigger, control, and background taxonomies. Due to the sheer number of classes in the FSD50K dataset, we again relied on an ChatGPT 5 to draft mappings between classes, prompting the LLM to map from one class to another if and only if the FSD50K or ESC-50 class is strictly linked to a class in our taxonomy (e.g., "chopping food" to "knife cutting"). We then reviewed each mapping to verify correctness.

As a result of this process, we identified 5 classes containing relevant trigger sounds from ESC-50 and 18 classes pertaining to trigger categories from FSD50K. Only FSD50K contained classes relevant to the control and background taxonomy. Because our control and background taxonomies and FSD50K followed the AudioSet ontology, such mappings were one-to-one. That is, we identified four control classes and 184 background classes from FSD50K. Table 1 shows the contribution of each dataset to the desired sound classes.

	Sound Class		
Dataset	Trigger	Control	Background
FSD50K	2167	894	28640
ESC-50	200	0	0
FOAMS	50	0	0

Table 1: Overview of Audio Used from Source Datasets.

We also examine the distribution of the trigger classes across the gathered samples, shown in Fig. 2. We see that the distribution of trigger samples is not uniform. Typing and plastic crumbling are the most common, with 434 and 429 instances, followed by human Breathing (352) and clearing throat (337). Water drops, chewing, and knife cutting are moderately represented. In contrast, basketball dribbling and flipping newspaper appear only five times each, with all samples originating from FOAMS. Neither the FSD50K or ESC-50 possess additional samples. We also note the existence of 28 trigger sounds with two labels, the most common pair being clearing throat and human breathing (18). A full description of multi-labelled samples can be found in Appendix A.2.

3.2.2 Train/Validation/Test Split

Ultimately we wish to use this misophonia-centric binaural dataset as the basis for an deep-learning ANC model. Therefore, we will have to split the binaural mixes into a train/val/test split. Given that the binaural sounds are a combination of source sounds, it is imperative that we apply the train/val/test split to the trigger, control, and background sounds before mixing. If we do not proceed in this manner, then a binaural mix in the test set may have the same trigger sound as a binaural mix in the training set. This would expose the ANC model to data leakage, the greatest offense in machin learning.

Given the high-quality audio and labels of the FOAMS dataset and the FSD50K evaluation set, we elected to dedicate all collected data from them to our test set. By doing so, we ensure a robust evaluation of the ANC model. This leaves us to split the ESC-50 and the FSD50K development set into train, validation, and test splits. Now, naively generating splits of these datasets would still run

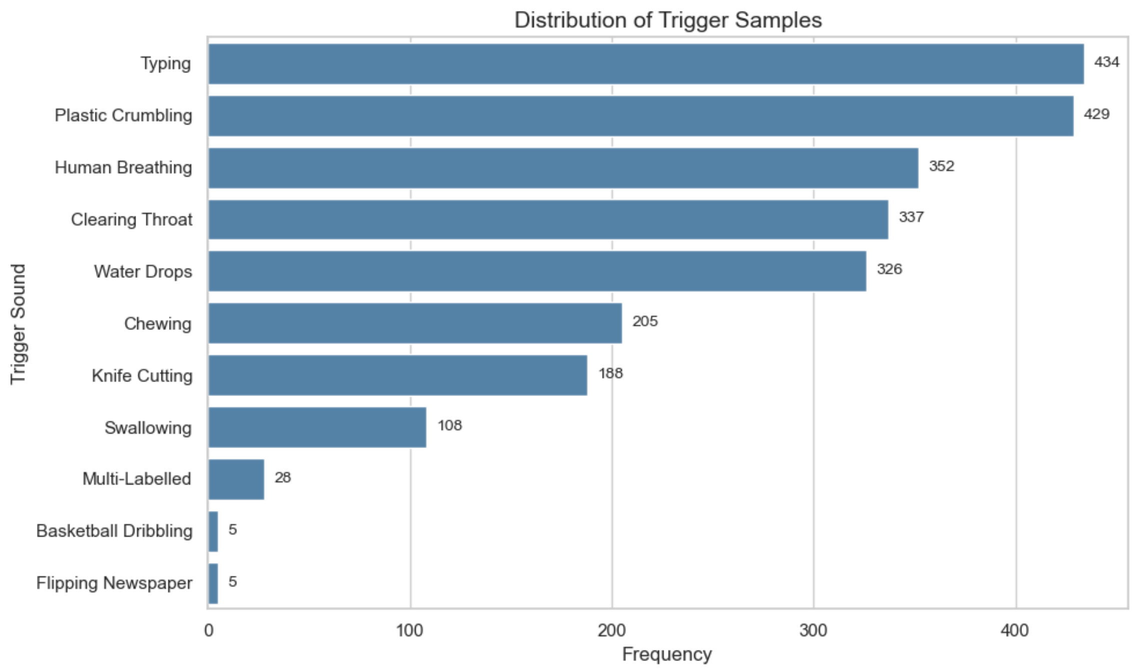

Figure 2: Distribution of Trigger Samples.

the risk of data leakage. This is a consequence of the fact that ESC-50, FSD50K, and UrbanSounds8K (the parent dataset of FOAMS) are subsets of the Freesound archive [2, 3, 13]. We must ensure that any duplicate sound events exist in the same split. Fortunately, all sound events across all datasets track the original Freesound IDs. With these, we develop a two-step splitting method to prevent data leakage. For any sound event, we first check if its Freesound ID coincides with the Freesound ID of any FOAMS or FSD50K evaluation data. If it does, we allocate it to the test set. If it does not, we then hash the Freesound ID, and place it in a split based on the resulting hash. The range of the hash function is divided according to a 70/20/10 split. This approach allows any sound with a globally unique id to be added to the source data in the future while respecting the existing splits. This does not guarantee that our source data is split exactly along those proportions, especially given that using FSD50K eval and FOAMS enforces an automatic lower bound on the test split size. Nonetheless, the prevention of data leakage and extensibility warrants such a tradeoff. Table 2 describes the exact size of each split across triggers, controls, and backgrounds.

	Sound Class		
Split	Trigger	Control	Background
Train	1646	675	21672
Validation	548	91	4021
Test	223	128	2947

Table 2: Number of examples in source data across splits and sound class.

3.3 Binaural Mixing

To synthesize our source audios into binaural soundtracks, we rely on Binamix, a recent python package for programmatic binaural mixing [22]. While not misophonia-specific, Binamix provides the technical substrate for generating paired scenes and controlling spatial factors needed by selective extraction models. Binaural audios are created by convolving a non-spatial audio with head-related impulse response (HRIR) or binaural room impulse response (BRIR) data. This generates two separate signals: one for each ear. In tandem, these two signals simulate spatially-located audio. For our purposes, BRIR data is particularly relevant. Binamix uses BRIR data from the SADIE II database [1] to convolve with non-spatial audio. The SADIE II database includes BRIR signals from 50 combinations of azimuth and elevation angles for 20 different subject heads (2 dummy, 18 human). This provides the means to simulate spatially-located audio for a diversity of different head configurations.

To generate the dataset, we binaurally mix one or more foreground sounds (trigger or control) with a series of background sounds. Our exact methodology is shown in Fig. 3. As the primary mixing function in Binamix expects soundtracks of the sample duration, we first pad the source audios such that they are all of equal length. For each audio that requires padding, We add a random duration of silence before and after. During this process, we also equalize the volume between the soundtracks using root mean square normalization (RMS): a standard for volume control [23]. Equipped with the padded and normalized audios, we then randomly configure the following parameters for the binaural mixing: subject head, azimuth and elevation of each soundtrack, the volume of the foreground audio, reverb effect, and the reverb type[2]. For each binaural mix, we use the BRIR data of the specified subject head, azimuth and elevation, as well as a fixed sampling rate of 44.1 kHz. Not only does these correspond to the sampling rate of Freesound audio files, but this is consistent with the sampling rate used by Veluri et al. [15] in their pipeline. We then apply Binamix's binaural mixing function on the foreground and background audios, using the randomly configured parameters. The resulting binaural mix, as well as the ground truth binaural audio, required for the ANC, are saved. When a trigger sound is mixed with background sounds, the binaural trigger sound acts as the ground truth. If a control sound is used for mixing, the ground truth is silence. This is because the ANC model should learn to avoid suppressing non-trigger sounds.

Because of the abundance of mixing parameter combinations, the binaural dataset can be greatly scaled despite the limited amount of source data. To support this, we release a Python pipeline that generates mixtures on the fly.

As mentioned, we release a *canonical* version of the dataset. This provides pre-computed mixtures that are computationally inexpensive to use during training and validation, and that can be combined

[2]Per Binamix, reverb type reflects different room settings for the acoustic environment. Options are: theater, office, small room, or meeting room

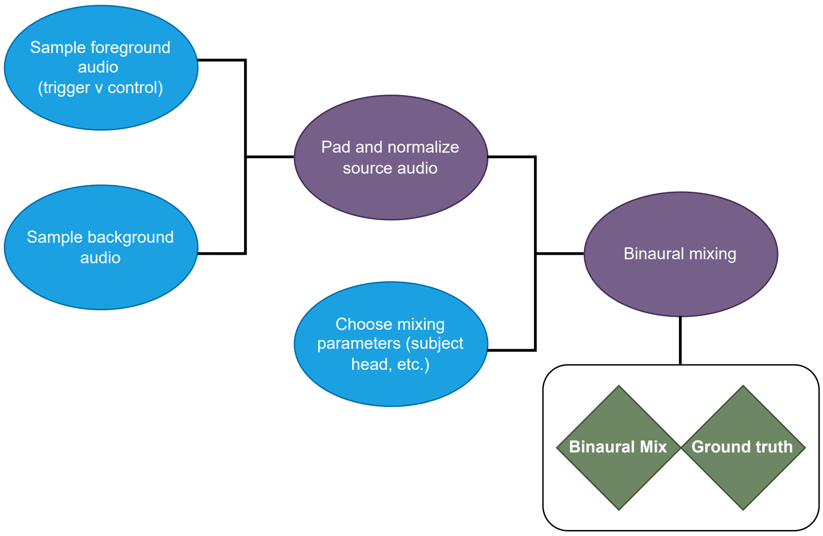

Figure 3: Binaural Mixing Pipeline. Blue represents random sampling process, purple deterministic functions, and green the output.

with on-the-fly mixtures when desired. Crucially, the canonical test split ensures that different models can be evaluated on an identical set of mixtures, enabling direct comparability across studies. We name it `canonical-v1` to anticipate that it could be improved in future versions. This version has one trigger per mixture and 1-3 backgrounds. It took 24 hours to generate[3] and contains 30,000 binaural mixes taking up 72 GB. The split sizes were chosen roughly proportional to the number of trigger samples in each split, yielding 3,000 samples (9 GB) in the test split, 7,000 samples (17 GB) in the validation split, and 20,000 (47 GB) in the train split.

For each mixture, we keep track of all associated metadata. This includes metadata of the foreground and background sounds, the parameters used to mix, and licensing information. In this way, the metadata files contain all information needed to reproduce the sounds (except for the sounds themselves), and can be used to determine, e.g., where in the mixture a specific trigger appears.

3.4 Licensing

Any open-access dataset must carefully consider the licensing that applies, as this impacts the uses of the dataset. The source data used in this study was sourced from publicly available datasets with clearly defined licenses. All sounds have different Creative Commons (CC) licenses, which we get from the source datasets' metadata and from the Freesound API. Each source dataset is also released on CC licenses: FSD50K under the CC BY 4.0 license, allowing redistribution with proper attribution. FOAMS and ESC-50 data is under the CC BY-NC 4.0 license, further restricting their use to non-commercial purposes. The SADIE II database is released under the Apache license. Data generated

[3]2.9 seconds per sample with 4 cores in parallel, or 11.4 seconds per sample with a single thread. Hardware used: Intel(R) Core(TM) i5-6600K CPU @ 3.50GHz (4 cores); 15Gi RAM; Ubuntu 22.04.5 LTS.

using our methodology is released under the CC BY 4.0 license and our code is released under the MIT license. We keep track of all licenses relevant for a given sound mixture and provide it as part of the metadata, in order for any user to be able to respect any constraints such as attribution or commercial use.

4 Experimental Validation Methodology

This section outlines how we plan to empirically validate that our *trigger* audio mixtures elicit higher discomfort than matched *control* mixtures. We describe how we prepare the study data, the study design and how we plan to analyze the obtained data. The protocol is fully specified here as a planned study; no data have been collected yet, due to waiting times for the ethics approval. A more detailed specification is provided in Appendix B.

Compared to prior misophonia-related ANC studies, our planned methodology for validation improves along four dimensions:

First, we provide stronger statistical inference by treating subjects and stimuli as random rather than fixed, using a crossed subject–by–item mixed-effects model to generalize beyond the specific clips and subjects included [24–26]. This avoids treating our particular stimuli or participants as fixed and reduces the risk of overestimating effects due to idiosyncratic clips or individuals.

Second, we conduct the experiment entirely online, without researcher presence, enabling recruitment of a larger and more heterogeneous sample than typical small-sample laboratory pilots. This design also mirrors the real-world use case of ANC systems deployed with everyday consumer headphones in unsupervised settings.

Third, our goal is not to establish formal clinical validation of the dataset, but to benchmark our sourced triggers against a small set of well-characterized reference stimuli. We leverage clinically validated trigger clips and explicitly compares them to newly sourced triggers, providing some indication of whether the broader dataset elicits discomfort patterns comparable to these reference stimuli. This means that the dataset will *not* be clinically validated, but it will be more closely aligned with existing clinical research than previous misophonia ANC research.

Fourth, we validate the *stimuli* rather than any specific ANC model. This separation of stimulus validation from model evaluation reduces the burden on future method papers, which can rely on pre-validated inputs. They should ideally still perform a separate evaluation of model *outputs*, and our experimental design can be reused for that purpose by substituting the control mixtures with mixtures that have been processed by the ANC system.

Note that the study has not yet been conducted due to waiting times for the ethics approval. The protocol described here should be pre-registered before any data collection begins to ensure transparency.

4.1 Preparation

For the validation study, we prepare a fixed set of 16 *item pairs* consisting of two *versions*: a trigger and a control. Within each pair, the trigger and control clips share the same background and other mixing configurations, differing only in foreground sound used.

For each of the 8 categories, we sample two trigger clips: one validated by FOAMS [4] (see Section 2) and one drawn from our broader library of non-FOAMS trigger recordings. If the category does not contain each of these, we exclude the category from the experiment. This pairing is chosen specifically to support a later FOAMS–equivalence analysis, and it allows us to cover all selected trigger categories for each subject while keeping the overall study length short enough to minimize dropout.

We sample equally many control clips. Matching the number of control clips to the number of trigger clips ensures that every background is paired with both a trigger and a control version, with the same mixing options. This balance enables us to estimate and control for the discomfort driven by the background sounds. It also limits the total number of potentially aversive trials per participant.

Each item pair is randomly assigned to item mask A or B, under the condition there must be an equal number of A- and B-pairs within each category. Upon entering the experiment, each subject is randomly assigned to group A or B using central counterbalancing targeting a 1:1 ratio. To avoid any subject hearing the same background more than once, Group A hears the trigger version of A-masked items and the control version of B-masked items, while group B hears the opposite. Thus, each background is presented approximately equally often with a trigger and with a control foreground. This mirrored design preserves between-subject counterbalancing while avoiding within-subject repetitions of the same background, which could otherwise introduce memory-based comparisons or demand characteristics.

To complement these mirrored between-subject comparisons, we also include a single *anchor item* designed to provide a secondary within-subject contrast. The anchor item consists of a shared background paired with both a control version and trigger of the chewing category, as this is the most common. Contrary to the non-anchor items, Group A hears the control version of the anchor as their first trial *and* the trigger version as their final trial, while group B hears the opposite order.

For the main analysis, ratings are only included for non-anchor items and the first shown item of the anchor pair, in order to minimize memory and carryover effects. This keeps the primary estimand focused on group-level differences between trigger and control clips.

4.2 Participants and ethics

Participants in the planned validation study will be adults (18+) with self-reported misophonia. Recruitment will follow established channels: Dr. Jane Gregory's research mailing list[4], outreach via universities and autism support centers, and a post on `/r/misophonia` following prior work [11]. Participants may also share the study link, enabling controlled snowball recruitment.

This approach will yield a convenience sample not representative of all individuals with misophonia, but it is appropriate for our aim of testing whether the curated mixtures elicit discomfort in a population of people who experience misophonia in everyday life. We will collect and report demographics information to help asses any biases.

Participation is voluntary and uncompensated. Informed consent will clearly state that participants may stop any sound immediately, withdraw at any point, and that no identifying information is collected. The consent form will include a brief screening step excluding individuals who report PTSD, past panic attacks triggered by everyday sounds, or other conditions that may elevate risk.

Expected risks are limited to transient discomfort, mitigated by the screening in the consent form, the stop button and immediate discontinuation option. We are inspired by procedures used safely in prior misophonia and sound-intolerance research [11, 19]. Risk is further reduced by the fact that approximately half of the trials present control mixtures

The protocol has *not* yet been approved by the IT University of Copenhagen's ethics committee. We have been in dialogue with the committee through our supervisor, and no data will be collected until formal approval is granted. The study is intended solely as a validation of the dataset and is not a diagnostic or clinical procedure.

4.3 Study Design

A summary of the end-to-end flow for a single subject is shown in Fig. 4.

Subjects are first introduced to the experiment, provide informed consent and demographic information (age group, gender and country).

Then, they indicate which of the trigger categories in our taxonomy they personally find aversive, followed by the Duke Misophonia Questionnaire (DMQ) [16] as described in Section 2. We administer only the *Symptom Severity* composite sub-scale, using it solely as a research screening measure to characterize self-reported misophonia-related symptoms in a time-efficient way. This scale captures core affective, physiological, and cognitive responses associated with misophonia but is not used for any form of clinical diagnosis or treatment decision. We chose this composite because it aligns closely with the dimensions of misophonic responding that are most relevant our discomfort-based validation.

[4]The authors have been in contact with Dr. Jane Gregory. See `https://soundslikemisophonia.com/recruiting`

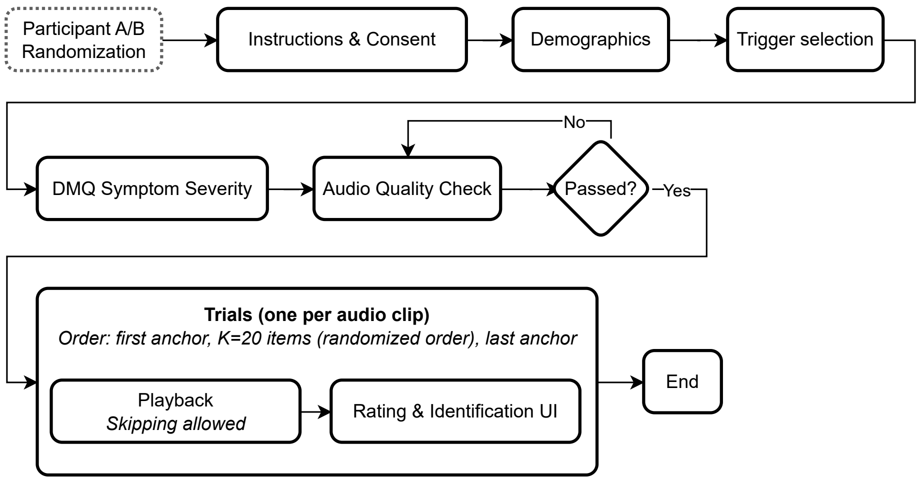

Figure 4: Flow of online validation experiment.

Before proceeding to the trials, the subject completes an *Audio Quality Check*. It is intended as a pragmatic check that participants are attending and that their playback setup can support basic recognition of everyday sounds, while remaining compatible with common consumer devices such as in-ear headsets. It is done by having the subject label some audio clips from a random subset FSD50K using a forced three choice selection. They must label 3 out of 4 correct to pass the test[5]. If they fail the check, the audio clips used to check are re-sampled and the test is presented again until they pass.

The trial phase consists of a sequence of audio mixtures. First, clear instructions are given. Then, they are exposed to the first anchor item (see Section 4.1). This is followed by the $K = 16$ non-anchor item pairs in randomized order. Finally, a second exposure to the anchor of the other version.

Each trial consist of a *Playback* stage followed by a *Rating & Identification* stage. In the Playback stage, the subject hears a brief washout segment of pink noise followed by the sound mixture. The brief pink-noise segment serves to standardize the acoustic context across trials and to reduce carry-over effects from the preceding mixture. Participants may stop the sound at any time if it becomes too uncomfortable.

Regardless whether they listened to the full sound, they then provide two responses: a discomfort rating and an identification of which trigger category/categories they identified. We adopt the same colored 0–5 Likert scale for discomfort as Hansen et al. [19] to facilitate informal comparison with prior work. The identification question functions as a manipulation check on whether participants perceived the intended category and supports secondary analyses of category recognizability.

The experiment has been implemented using jsPsych [28] and JATOS [29].

[5] We also considered using a Huggins Pitch headphone check, such as Milne et al. [27], but did not do it, since we found it to exclude users of in-ear headsets.

4.4 Planned Analysis

Note that no data has been obtained yet, but this section describes how such data should be analyzed.

The analysis quantifies how much the trigger audio clips increase discomfort relative to matched controls, both for declared trigger categories (personalized effect) and across all triggers (global effect). Full technical details are given in Appendix B.2.1; here we summarize the main ideas and assumptions.

Each trial is defined by a subject s, an item pair i, and a version $v \in \{\texttt{ctrl}, \texttt{trig}\}$, together with the rating r and the participant's category judgments. From the design, we derive indicator variables such as whether the clip contains any trigger sound (isTrig), whether a trigger belongs to one of the participant's self-declared trigger categories ($\mathrm{isDeclaredTrig}$), whether the trigger was sourced from FOAMS ($\mathrm{isFOAMS}$), and a centered order index (Order) capturing potential fatigue or habituation.

We exclude trials with rating response time of over 90 seconds from the primary analysis, as well as the second exposures of the anchor item. At the subject level, we include only participants whose DMQ Symptom Severity score exceeds the pre-specified cutoff recommended by Rosenthal et al. [16] and who contribute at least 3 included trials, ensuring that subject-level random effects are based on more than isolated single observations.

We analyze ratings using a linear mixed-effects model (LMM) with crossed random effects for subjects and items [e.g., 24–26]. This structure allows us to generalize beyond the specific participants and audio mixtures in the study, under the assumption that they are reasonably representative samples of sounds and populations of individuals with misophonia. We use a linear rather than ordinal specification because it yields effect estimates directly in rating units and is supported by evidence that parametric models are robust for Likert-type data in practice [30], as well as by prior work in this area [e.g., 19].

In lme4 notation [31], our model is:

$$
\begin{aligned}
\text{rating} \sim\ & \underbrace{\text{isTrig} + \text{isDeclaredTrig} + \text{isFOAMS} + \text{Order}}_{\text{fixed effects}} \\
& + \underbrace{(1 + \text{isTrig} \,||\, s) + (1 + \text{isTrig} \,||\, i)}_{\text{random effects for subject and item}}
\end{aligned}
$$

From the fitted fixed effects, we construct two estimands:

$$
\begin{aligned}
\Delta^{\text{personal}} &= \beta_{\text{isTrig}} + \beta_{\text{isDeclaredTrig}} + \phi \cdot \beta_{\text{isFOAMS}} \\
\Delta^{\text{global}} &= \beta_{\text{isTrig}} + \hat{\psi} \cdot \beta_{\text{isDeclaredTrig}} + \phi \cdot \beta_{\text{isFOAMS}}
\end{aligned}
$$

where ϕ is the proportion of triggers that are FOAMS-sourced in the complete dataset, and $\hat{\psi}$ is the empirical share of trigger trials belonging to a subject's declared trigger categories in the analyzed sample.

The primary estimand is Δ^{personal}, summarizes the expected difference in discomfort between a typical control clip and a trigger clip from one of the participant's own declared trigger categories. This quantity is most relevant for personalized ANC systems that aim to suppress only user-specific triggers. The secondary estimand Δ^{global} summarizes the average discomfort difference between trigger and control clips across all categories and participants, irrespective of the subjects' declared trigger categories.

For both estimands, we will report point estimates together with 95% confidence intervals and two-sided p-values with $\alpha = 0.05$. If the confidence interval does not include zero, we interpret this as evidence that trigger mixtures are statistically significantly more triggering than the controls. Beyond statistical significance, we interpret the magnitude of the estimated effect using the practical thresholds motivated in Appendix B.1: differences smaller than 0.25 points are interpreted with caution and are not, by themselves, taken as clearly practically meaningful, whereas effects of 0.6 points or more are viewed as strong evidence that the mixtures are robustly triggering.

Our interpretation of intermediate effect sizes is further informed by a FOAMS-equivalence analysis. Since FOAMS [4] contains clinically validated trigger sounds, we are interested in determining whether non-FOAMS triggers produce practically equivalent discomfort ratings. This would be an indication of validity for our methodology for sourcing trigger sounds from existing, non-validated datasets. Such an analysis is described in more detail in Appendix B.2.4. We emphasize that this comparison is used as a practical benchmark rather than as a substitute for formal clinical validation of our stimuli.

Although one could, in principle, extend this specification with category-specific fixed effects[6], we deliberately avoid such a saturated structure in the primary model. Each of the categories is represented by only two item pairs, and the A/B masking ensures that a given subject typically contributes at most one trigger trial per category. Under these constraints, a category-by-trigger parameterization would be high-dimensional relative to the available data and would yield unstable, weakly constrained estimates, particularly for interactions involving `isDeclaredTrig`. We therefore retain a collapsed fixed-effect structure not dependent on the category for the primary analysis, and treat per-category summaries as secondary analyses. This implies that Δ^{personal} and Δ^{global} are category-averaged quantities.

Furthermore, Appendix B.2.4 describes a set of secondary and descriptive analyses. These include: summarizing how often participants correctly identify the intended category; exploring cor-

[6]For example, by replacing `isTrig` with a factor `trigCategory` and modeling terms such as `trigCategory` and `trigCategory` × `isDeclaredTrig` to obtain separate trigger and personalization effects for each category.

relations between DMQ Symptom Severity and average trigger ratings; reporting per-category mean ratings; and anchor-pair within-subject differences.

Overall, the analysis pipeline rests on several explicit assumptions: the DMQ Symptom Severity composite is a suitable proxy for clinically meaningful misophonia symptomatology in an online sample; our sample of participants and sounds can be treated as sufficiently diverse and typical for the purposes of mixed-effects generalization; and that Likert ratings can be modeled as approximately continuous (see Norman [30]). Accepting these assumptions yields that the study is an interpretable, statistically grounded validation of the dataset's ability to elicit differential discomfort, supporting its use as a benchmark for misophonia-focused ANC research.

The analysis script was implemented in R using lme4 [31] and tested using dummy data generated with Python.

5 Discussion

5.1 Design Decisions

Our pipeline is designed to make incorporating new source datasets straightforward. This is a design choice we established from the start, given the continuously growing field of misophonia research. We wish to be able to draw from any relevant audio data with little friction and without affecting our existing misophonia dataset. For instance, during this project, Oh et al. [17] released a preprint introducing a well suited candidate for an additional source of trigger sounds: an audio–visual corpus of misophonia trigger clips. Concretely, our implementation of sampling, bookkeeping and mixing work against an abstract source-data interface, allowing this and other future datasets to be plugged into the same taxonomy and mixing pipeline with minimal code changes. And the splitting logic is defined at the level of global identifiers rather than specific datasets, so newly added sources can be assigned to train/validation/test without impeding on existing splits.

On the other hand, some design decisions were made that could be candidates for change. For example, our canonical dataset only includes mixes with one trigger sample, as opposed to multiple. The rationale behind this is tied the ground truth audio which we would like an ANC model to extract. Our algorithm outputs the binaural trigger as the ground truth. Mixing with multiple triggers would result in a binaural ground truth containing all the triggers. Because misophonia triggers are idiosyncratic, it may be the case that a patient is highly sensitive to plastic crumbling sounds, but unaffected by the other classes. If said patient wished to train a model solely for their needs, then all ground truth audio should either be plastic crumpling, or silence. If we had included several triggers in each mix, then our dataset would not be suited for this task. Our current canonical dataset can be easily adapted however, by filtering on mixes containing only plastic crumpling. Now, if we consider a patient that is severely disturbed by all trigger classes, then a model that can suppress all trigger sounds, even if they occur simultaneously, would be required. In this case, the model should be trained on data with mixes and ground truths that incorporate several triggers. While our canonical dataset is not adapted for this, our pipeline is.

Another design decision that could be modified is the choice of sampling rate for the binaural mixes. The `canonical-v1` dataset uses a fixed sampling rate of 44.1kH. This value is aligned with the sampling rate of Freesound audio and the binuaral mixes of Semantic Hearing. One could argue that such a high sample rate is unjustified, given that none of the trigger classes appear to occur at high frequencies. Moreover, sampling at lower frequencies has practical effects, as it reduces the data throughput to the ANC model. Admittedly, we did not investigate the frequencies present in the source trigger sounds. This is certainly a limitation of the EDA we performed on the source data. That being said, the sampling rate of our mixing algorithm can be easily be configured. In future work, if it becomes apparent that 44.1kH is unnecessary, users can generate new audio at a lower sampling rate with little effort.

5.2 Future Work

We propose that the future development of misophonia-focused Active Noise Control (ANC) should be guided by three pillars: *Cross-disciplinary integration*; *verified triggers and clinically inspired ontology*; *integrated, patient-centric control*.

5.2.1 Cross-disciplinary Integration

Development should be a collaboration between psychology/clinical science and machine learning, ensuring that datasets, tasks, and evaluations reflect clinical constructs relevant to misophonia.

Progress on misophonia-focused machine learning will require close coordination between psychological science and computational modeling. Our project has taken initial steps by grounding the dataset design in constructs drawn from clinical research on misophonia trigger sounds. We also incorporated survey measures aligned with existing clinical screening tools to support the eventual data analysis. The ideal next step stage is to integrate clinicians and behavioral researchers into the development loop to formalize this work. This would include: selecting metrics that reflect clinically meaningful change and map model performance onto real symptom profiles. These would ensure that an ANC model performs well quantitatively, but more importantly, truly improves the quality of life of users.

5.2.2 Verified Triggers and Clinically Inspired Ontology

Audio classes should be grounded in a common, clinically anchored taxonomy, and the stimuli used should be empirically verified to trigger individuals with misophonia and compared with clinically verified stimuli.

Our ontology consolidates heterogeneous trigger categories from FOAMS and Claiborn et al. [20] into the Duke Misophonia Questionnaire (DMQ) taxonomy. This consolidation ensures that our dataset aligns with a clinically motivated backbone rather than ad hoc or crowdsourced labels. However, our ontology is limited to these three publications. Future work should conducting a systematic literature review of existing misophonia taxonomies to formalize a complete taxonomy. In parallel, researchers should develop a clinically verified control taxonomy that specifies non-trigger sounds. Our work directly refers to the control sounds motivated by Hansen et al. [19]. But there has yet to be a thorough investigation of non-triggering sounds. Together, these efforts would yield a rigorously defined, clinically anchored ontology for both misophonia trigger and non-trigger sounds.

5.2.3 Integrated Patient-centric Control

We see the following components as instrumental for the path for future misophonia ANC: classification to label trigger categories; spatial extraction/suppression to remove them while preserving non-triggers; and biosignal monitoring during validated exposures to capture user state for personalization and evaluation.

Our dataset is intended primarily for training selective Active Noise Control models. But its structure also enables future work on physiological responses to misophonia triggers. Because the clips are binaural, labeled, and designed to reflect real-world acoustic conditions, they provide a controlled stimulus set that can later be used to examine how different sounds modulate biosignals such as heart rate variability, electrodermal activity, and respiration [32–34].

How does work towards patient-centric control? Consider a wearable device or mobile app that monitors biosignals passively while an embedded classifier, developed from our dataset, detects likely trigger events in the acoustic environment. When such an event is detected, the system captures a short window of biosignals. Over time, this yields a personalized corpus of paired acoustic event and physiological response samples without requiring continuous annotation by the user, This could lead to downstream models to track individualized severity of reaction, or refine ANC systems through user-specific physiological feedback. Although purely theoretical at this point, these examples illustrate how our current dataset could serve as as a starting point for user-centric misophonia research.

Furthermore, evaluation metrics must reflect the asymmetric risks inherent in misophonia mitigation. Because individuals differ in both trigger profiles and reaction severity, models should be assessed with metrics that weight false positives and false negatives according to their practical impact. False negatives may allow aversive sounds to pass through, while false positives may suppress desired audio or degrade listening quality. The cost of false positives or false negatives will depend from user to user. Future research will be required to take a patient-centric approach to accommodate the variety of patient profiles.

6 Conclusion

The broader aim of research in this area is to create tools that genuinely reduce the burden misophonia places on everyday life. Individuals with misophonia must often navigate ordinary settings while managing sudden, intense reactions to specific sounds, yet they have few effective strategies for maintaining comfort and control. Developing meaningful assistance is therefore a complex, long-term effort that demands sustained cross-disciplinary collaboration. Prior research has been essential in identifying common triggers and characterizing the severity of associated discomfort, clarifying the scope and challenges of the problem. With this foundation in place, the field can now move toward building practical technologies that address these needs. Our work intends to do exactly this. We summarize our contributions below:

- *Framework.* We have described a principled framework inspired by machine learning and clinical research, working towards the three pillars mentioned in Section 5 that align data generation, taxonomy, evaluation and real-world usage.

- *Dataset.* We have released the first large-scale, open-access binaural corpus of mixed audio clips with trigger/control foregrounds, designed for selective misophonia-focused ANC. We release a pre-made canonical version of the dataset, as well as a extensible pipeline to generate binaural mixtures on the fly.

- *Experimental design.* We have designed and implemented an online experiment for empirical verification of trigger effects on patients via discomfort ratings in screened participants. We were, however, not able to conduct the experiment yet due to waiting times for the ethics approval.

Together, the framework and dataset provide essential building blocks for developing selective ANC models tailored to misophonia. The framework defines how taxonomy, data generation, and evaluation criteria align with real-world needs, while the dataset supplies the large-scale, structured audio material required to train and benchmark such systems. In parallel, the experimental design offers a complementary foundation for future research. Once implemented, it will provide the empirical validation of trigger effects

Building on this foundation, we aim to develop robust selective ANC systems and extend them toward multimodal, personalized interventions that address the heterogeneous needs of individuals with misophonia.

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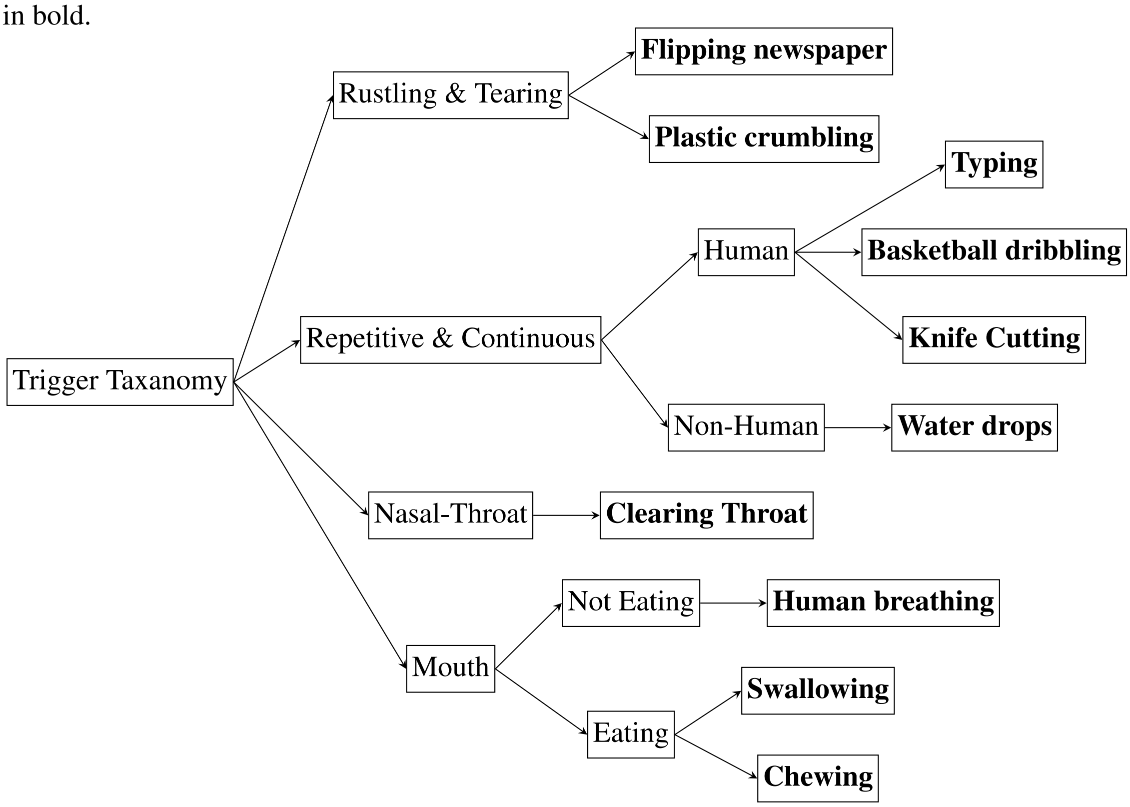
Appendices

A Extended Details Regarding Dataset Curation

A.1 FOAMS Trigger Classes within the DMQ

The tree below demonstrates how the FOAMS trigger classes fit within the DMQ taxonomy. FOAMS classes are in bold.



A.2 Distribution of Triggers within Multiclass Samples

Labels	Count
(Clearing throat, Human breathing)	18
(Chewing, Human breathing)	4
(Chewing, Clearing throat)	2
(Swallowing, Water drops)	2
(Chewing, Plastic crumpling)	2

Table 3: Counts per Multiclass Trigger Sample

B Extended Details Regarding the Experimental Validation Methodology

Overview. The study measures discomfort caused by misophonia trigger sounds relative to matched control versions in a between-subjects (*A/B*) mirroring design with a universal anchor pair. All subjects are exposed to the full stimulus set; the subject- and item-level A/B masks determine whether the trigger or control version is presented on a given trial. Personalization is applied *post hoc* during analysis by conditioning on each subject's declared trigger categories $T_s \subseteq \mathcal{C}$ (Appendix B.2.1). Subjects are indexed by $s = 1, \ldots, N$; item pairs by $i = 1, \ldots, K$; categories by $c \in \mathcal{C}$. Each item pair i has two versions $v \in \{\text{ctrl}, \text{trig}\}$.

Preregistration. This document will be preregistered on the Center for Open Science (OSF).

Privacy and sharing. Collect minimal, de-identified data; public release of anonymized datasets and code is intended (with informed consent). We only publish the data if 10 or more subjects participate. Any guidelines of the IT University of Copenhagen will be followed, and the independent ethics body will be consulted before collecting subject data.

B.1 Study Design

B.1.1 Participants

- **Enrollment.** We record responses from all subjects who (i) consent; and (ii) are 18 years old or above. (Appendix B.1.3). Analytic inclusion and exclusion are defined post hoc in Appendix B.2.1.
- **Subject Group.** Subjects are assigned to group $g_s \in \{\mathrm{A}, \mathrm{B}\}$ uniformly at random with a target 1:1 ratio, using a central balancer. However, the ratio cannot be guaranteed due to some participants being excluded or not completing the study.
- **Demographics.** We record age group, gender, and country.

B.1.2 Stimuli

- **Taxonomy.** We use $|\mathcal{C}| = 8$ trigger categories (flat, non-overlapping).
- **Item Pairs.** An *item pair* always has the same background sound; but has two *versions*, one where the background is mixed with a control (`ctrl`) and one mixed with a trigger (`trig`). To get $K = 2|C|$ item pairs by:
  - sampling K background sounds at random;
  - sampling K control sounds at random;
  - sampling 2 triggers for each category $c \in \mathcal{C}$, such that one trigger is sourced from FOAMS [4] and the trigger from non-FOAMS material.
- **Anchor.** A universal *anchor pair* $i^\star$ is presented to in both versions to all subjects in addition to the non-anchor trials. It is of category chewing, since this is the most common trigger. It is useful for the secondary analysis described in Appendix B.2.4. Group A hears $i^\star_{\text{ctrl}}$ as the first trial and $i^\star_{\text{trig}}$ as the final trial; Group B hears the reverse. The second exposure is excluded from the analysis set (see Appendix B.2.1).
- **Item Mask and Display Rule.** Randomly assign each non-anchor pair to mask $g_i \in \{\mathrm{A}, \mathrm{B}\}$, such that there are equally many A and B pairs within each category. All subjects are exposed to the same set of item pairs. For non-anchor item pair i, the version subject s is exposed to is determined by:

$$\text{version}(s, i) = \begin{cases} \texttt{trig}, & \text{if } g_s = g_i, \\ \texttt{ctrl}, & \text{if } g_s \neq g_i. \end{cases}$$

- **Stimuli size.** Therefore, we sample a total of $K+1 = 17$ pairs, and each subject is exposed to $K+2 = 10$ sounds if they complete all trials.
- **Order.** For each subject, all non-anchor item pairs are presented in a uniform random order; the per-subject order index is saved.

B.1.3 Trial Flow and User Interface

Each subject that fully participates in the study will complete this process:

1. **Introduction and Consent.** Introduce the study to the subject and collect consent for their participation.
2. **Demographics.** Collect demographic information.
3. **Trigger selection.** The subject selects which of the trigger categories trigger them. They select from the taxonomy C with the option to specify "*Others*" in free text and choose "*None of the above*".
4. **Questionnaire.** The subject answers the questions relevant to the Symptom Severity sub-scale of the Duke Misophonia Questionnaire (DMQ) [16].
5. **Audio Quality check.** The subject label some audio clips from a random subset FSD50K using a forced three choice selection. They must label 3 out of 4 correct to pass the test. If they fail the check, the audio clips used to check are re-sampled and the test is presented again until they pass.
6. **Trials.** For each item i to be rated by the subject (including the anchors):
   (a) **Playback.** The screen shows an icon illustrating that sound is playing. Meanwhile, play a fixed washout consisting of a constant $5\,\mathrm{s}$ low-level pink-noise followed by $0.5\,\mathrm{s}$ silence. Then the item is played. A button is always available: *"Sound is too uncomfortable, stop playback."* Pressing it immediately stops playback, records the time listened, and proceeds to the rating UI (ratings are still included in the study).
   (b) **Rating UI.** After sound playback, collect information in a similar fashion to Hansen et al. [19]:
      - **Rating.** A Likert scale with discrete values $\{0, 1, 2, 3, 4, 5\}$, the same colors Hansen et al. [19] used, and title "Rate the discomfort/anxiety you felt while hearing the sound".
      - **Identification.** A question, "Which of the following categories did the sound you just heard belong to", with options being a multiple choice checklist of the trigger categories C as well as *"None of the above"*.
7. **End.** A text informs the subject that they are done with taking the study. An option for sharing the link is available.

B.1.4 Safety and Ethics

The subjects always know when a new sound is presented to them, and choose when to proceed to listening to that sound. Subjects control exposure via the stop button and may discontinue participation at any time.

Hansen et al. [19] and Wunrow [11] have conducted similar studies, approved by ethics committees. Based on these studies, we expect to observe discomfort ratings below 4 out of 5, even for the most triggering categories. Half of the sounds presented to participants are controls, limiting the number of sounds we expect to trigger them.

We believe the dataset will greatly contribute to the development of applications that will meaningfully improve the lives of misophoniacs, namely noise canceling headphones.

B.2 Analysis Methodology

B.2.1 Analysis Populations and Data Preparation

Variables. Each trial consist of:

- a subject s who said they are triggered by T_s categories;
- one item i of version $v \in \{\texttt{ctrl}, \texttt{trig}\}$;

- the category $c(i,v) = \begin{cases} \texttt{ctrl}, & \text{if } v = \texttt{ctrl}, \\ \text{the category of the trigger}, & \text{otherwise} \end{cases}$
- the rating $r_{s,i,v} \in \{0,1,2,3,4,5\}$; and
- the identified categories $I_{s,i}$.

We then construct variables:

$$\begin{aligned}
\mathrm{isTrig}_v =\ & \mathbf{1}[v = \text{trig}] \\
& \text{(i.e., contains a sound of some trigger category)} \\
\\
\mathrm{isDeclaredTrig}_{s,i,v} =\ & \mathbf{1}[v = \text{trig} \ \wedge \ c(i,v) \in T_s] \\
& \text{(i.e., contains a sound of a category the subject declared triggering)} \\
\\
\mathrm{Order}_{s,i} =\ & o - \frac{M+1}{2} \\
& \text{where } o \in \{1,\dots,M\} \text{ is the within-subject trial index for item } i \\
& \text{and } M \text{ the total number of planned trials} \\
& \text{(i.e., the order centered by planned study flow)} \\
\\
\mathrm{isFOAMS}_{i,v} =\ & \mathbf{1}[v = \text{trig} \ \wedge \ \text{trigger sound of } i \text{ is sourced from FOAMS}] \\
\\
\mathrm{isReused}_{s,i} =\ & \mathbf{1}[\text{the subject was previously exposed to some version of } i] \\
& \text{(only holds true for the second anchor item } i^\star\text{)}
\end{aligned}$$

Trial inclusion criteria. For the primary analysis, we only include trials (i) that were completed; (ii) the response was submitted within 90 seconds of the end of stimuli presentation; and (iii) $\mathrm{isReused}_{s,i} = 0$.

Subject inclusion criteria. For the primary analysis, we only include data from subjects who (i) exceed the pre-specified cutoff (mean ≥ 1.8) of the Symptom Severity composite sub-scale of the DMQ [16]; and (ii) where at least 3 trials were conducted and included.

B.2.2 Primary Model and Estimand

Model. Let r_{siv} be the rating of a trial. We model the data using Linear Mixed Modelling (LMM) following [e.g., 24–26]. In lme4 notation [31], our model is:

$$\begin{aligned}
\text{rating} \sim\ & \underbrace{\text{isTrig} + \text{isDeclaredTrig} + \text{isFOAMS} + \text{Order}}_{\text{fixed effects}} \\
& + \underbrace{(1 + \text{isTrig} \,||\, s) + (1 + \text{isTrig} \,||\, i)}_{\text{random effects for subject and item}}
\end{aligned}$$

Let $\beta_{\text{Intercept}}$ be the estimated intercept of the model, i.e., the model's baseline sound discomfort:

$$\beta_{\text{Intercept}} = E[r \mid \text{isTrig} = 0, \text{isDeclaredTrig} = 0, \text{Order} = 0, \text{isFOAMS} = 0]$$

For some fixed effect e, the estimated slope coefficient β_e represents the additive change in expected rating from a one-unit increase in e (for indicators, activating those indicators), holding all other predictors at the baseline. Some predictors are structurally dependent on isTrig, so for these the relevant baseline is $\text{isTrig} = 1$. Therefore, we interpret the coefficients as:

$$\begin{aligned}\beta_{\text{isTrig}} &= E[r \mid \text{isTrig} = 1,\ \cdots] - \beta_{\text{Intercept}}\\ &\quad \text{(i.e., global triggering effect)}\end{aligned}$$

$$\begin{aligned}\beta_{\text{isDeclaredTrig}} &= E[r \mid \text{isTrig} = 1,\ \text{isDeclaredTrig} = 1,\ \cdots] - \beta_{\text{isTrig}}\\ &\quad \text{(i.e., additional effect of having declared the trigger category)}\end{aligned}$$

$$\begin{aligned}\beta_{\text{isFOAMS}} &= E[r \mid \text{isTrig} = 1,\ \text{isFOAMS} = 1,\ \cdots] - \beta_{\text{isTrig}}\\ &\quad \text{(i.e., effect of having trigger sound come from FOAMS)}\end{aligned}$$

$$\begin{aligned}\beta_{\text{Order}} &= E[r \mid \text{Order} = 0\ \cdots] - \beta_{\text{Intercept}}\\ &\quad \text{(i.e., fatigue effect relative to the planned middle trial)}\end{aligned}$$

where "$\cdots$" denotes the remaining fixed effects at 0.

Personalized Estimand (Primary). The primary estimand is the observed triggering effect for declared categories, i.e. comparing typical trigger–declared trials to typical controls, using the fitted fixed effects β. We post-stratify to account for the oversampling of FOAMS-triggers.

Formally, we let

$$\Delta^{\text{personal}} = \beta_{\text{isTrig}} + \beta_{\text{isDeclaredTrig}} + \phi \cdot \beta_{\text{isFOAMS}}$$

where ϕ is the proportion of triggers that are FOAMS-sourced in the complete dataset.

Global Estimand (Secondary). The secondary estimand is the observed triggering effect irrespective of declared categories, i.e. comparing typical trigger trials to typical controls, using the fitted fixed effects β.

Formally, we let

$$\Delta^{\text{global}} = \beta_{\text{isTrig}} + \hat{\psi} \cdot \beta_{\text{isDeclaredTrig}} + \phi \cdot \beta_{\text{isFOAMS}}$$

where ϕ is the proportion of triggers that are FOAMS-sourced in the complete dataset, and

$$\hat{\psi} = \frac{\sum_{(s,i)} \mathbf{1}\left[\text{isTrig}_{s,i} = 1 \wedge \text{isDeclaredTrig}_{s,i} = 1\right]}{\sum_{(s,i)} \mathbf{1}\left[\text{isTrig}_{s,i} = 1\right]}$$

is the empirical share of trigger trials belonging to a subject's declared trigger categories in the analyzed sample.

Justification and rationale. This structure supports generalization to new subjects, item pairs, and categories. The crossed subject-by-item framework with random slopes remains essential to generalize beyond specific people and stimuli and to avoid stimulus-as-fixed-effect errors [e.g., 24–26]. Although ratings are on a bounded Likert-type scale (0–5), parametric analyses are well established as robust for such data; linear models are acceptable and typically recover correct inferences under wide conditions [30].

Since $\hat{\psi}$ is estimated from a relatively large number of trials, its sampling variance is negligible relative to the uncertainty in the fixed-effect coefficients. Accordingly, we treat $\hat{\psi}$ as fixed when constructing confidence intervals and p-values for Δ^{global}.

B.2.3 Estimation, Inference, and Decision Rules

Fitting. Estimate by restricted maximum likelihood (REML) for variance components.

p-values and intervals. We will report 95% confidence intervals and two-sided p-values for β using Satterthwaite small-sample degrees of freedom, consistent with standard mixed-model practice [26].

Interpretation of effect size. On the 0–5 discomfort scale we regard a difference of 0.25 points as the smallest effect that is *practically* meaningful. This corresponds to 5% of the full rating range and roughly a small shift relative to within-condition variability reported in Hansen et al. [19], where group-level contrasts between misophonia and control participants are typically in the 0.6 to 1.2 point range. In our more heterogeneous online setting with complex mixtures, we do not expect effects of that magnitude for all categories, but we take 0.25 as a conservative lower bound for a change likely to be noticeable across repeated exposures. Effects of 0.6 points or larger will be interpreted as strong evidence that the mixtures are robustly, meaningfully triggering. Our interpretation of observed effect sizes is further informed by the FOAMS-equivalence analysis described in Appendix B.2.4.

Sample Size Interpretation. No prospective power simulations are planned. We interpret results according to realized N: treat as *confirmatory* if $N \geq 50$; treat as *exploratory with caution* if $30 \leq N < 50$; and treat as *pilot* if $N < 30$. These thresholds reflect pragmatic guidance on stability and power for crossed LMMs with random slopes [e.g., 26].

Unequal contributions and missingness. The LMM naturally accommodates missing trials (timeouts) without imputation; all available trials contribute under MAR-type assumptions [e.g., 24, 26]. Order effects are modeled via Order.

Convergence and Fallback Strategy. If the model fails to converge, step down the random-effects structure in this order (documenting each change):

1. Change subject random effect from $(1 + \text{isTrig} \,||\, s)$ to $(1 \,|\, s)$
2. Change item random effect from $(1 + \text{isTrig} \,||\, i)$ to $(1 \,|\, i)$.
3. Change item random effect from $(1 \,|\, i)$ to a category random effect $(1 \,|\, c(i, v))$.
4. Drop Order fixed effect variable.
5. Drop isFOAMS fixed effect variable.

Re-evaluate convergence after each step; refit the maximal feasible model. This follows recommendations to include the richest justified random structure while maintaining estimability [25, 26].

B.2.4 Secondary and Descriptive Analyses

FOAMS equivalence We test whether trigger sounds sourced from FOAMS [4] yield practically equivalent discomfort relative to non-FOAMS triggers. We define the smallest effect size of interest (SESOI) as $\delta = 0.25$. Using TOST (Two One-Sided Tests) at $\alpha = 0.05$ with Satterthwaite df, we will conclude practical equivalence only if the 90% CI for β_{isFOAMS} lies entirely within $(-\delta, +\delta)$.

Anchor analysis. For each subject who rated both versions of the anchor item, compute the within-subject paired difference, $d_s = r_s(i^\star_{\text{trig}}) - r_s(i^\star_{\text{ctrl}})$. Report the mean $\bar{d}$ and 95% t-based CI across subjects.

Version balance. Monitor item-level version coverage; if any item pair's eligible exposures exceed a 75/25 trigger–control imbalance, report the imbalance with no analytic adjustments.

Skipped after. Trials with “Stop playing sound” pressed remain included; record `skipped_after` (seconds listened, or `None`) and report counts and proportions split by isTrig.

Identifications. Overall and split by category, report the proportion of trials where:

- the subject chose "*None of the above*" when identifying the sound, i.e. $|I_{s,i}| = 0$;
- the subject chose exactly the expected category, matching the dataset label, i.e. $I_{s,i} = \{c(i, v)\}$;
- the subject chose the expected category but also some unexpected, i.e. $\{c(i, v)\} \subset I_{s,i}$; and
- the subject did not chose the expected category but chose an unexpected, i.e. $c(i, v) \notin I_{s,i}$.

DMQ severity correlation. On the dataset, but where subjects are not excluded based on DMQ score, calculate the correlation between the DMQ score and the rating of trigger sounds.

Per-category effects. Report the average rating per category.

B.2.5 Reporting

Report mean, 95% CI and p-value for all fixed effects coefficients β as well as estimands, Δ^{personal} and Δ^{global}.

Report standard deviation for all random effects and residuals.

Provide convergence status, any fallback steps, and descriptive summaries.

B.2.6 Software

Analyses will be conducted in Python and R. The core model and inference follow standard mixed-model practice [e.g., 24–26].

C Use of Artificial Intelligence

We have done this project with the additional, partial help of Generative Artificial Intelligence (GAI) tools. Specifically, we have used GitHub Copilot (Gemini 2.5 Pro, Gemini 3.0 Pro, Claude 4 Sonnet, GPT-5, GPT-5.1) and ChatGPT (GPT-5, GPT-5.1). We have used these tools with assisting us in:

- programming and learning programming tools;
- writing and improving the language for this report;
- searching existing literature; and
- brain storming ideas.

All ideas and final decisions are our own. We have critically evaluated and, where necessary, independently verified any outputs produced by GAI tools, and we take full responsibility for the content included in this report. The tools were used as supportive aids rather than authoritative sources, and their outputs were not adopted uncritically or verbatim.